

Data Sciences Mini Project

Final Project Document

Phase 4 - Final Project Implementation

Electricity Consumption Prediction

Forecast electricity usage for households

Project Phase:	Phase 4 - Final Implementation
Target Variable:	PowerConsumption_Zone1
Final Model:	Random Forest Regressor
R2 Score:	0.5738
Dataset:	powerconsumption.csv (52,416 rows)

1. Final Model

The final model selected for this project is the Random Forest Regressor. This decision was based on the comparison of three regression models carried out in Phase 3 - Linear Regression, Random Forest Regression, and Gradient Boosting Regression. Among the three, Random Forest performed best on all three evaluation metrics and was therefore chosen as the final model for Phase 4.

Final Model Configuration

Parameter	Value
Algorithm	Random Forest Regressor
n_estimators	100
random_state	42
Training Data	80% (41,932 rows)
Testing Data	20% (10,484 rows)

Input Features

- Temperature - Temperature recorded at that time
- Humidity - Relative humidity
- WindSpeed - Wind speed
- GeneralDiffuseFlows - General diffuse solar radiation
- DiffuseFlows - Diffuse solar radiation

Target Variable: PowerConsumption_Zone1

2. Complete Python Source Code

train_model.py

This file handles the complete machine learning pipeline. It loads the dataset, selects the five input features, splits the data 80/20, trains the Random Forest Regressor, evaluates it using MAE/RMSE/R2, and saves the trained model as model.pkl.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import joblib

df = pd.read_csv('powerconsumption.csv')
features = ['Temperature', 'Humidity', 'WindSpeed', 'GeneralDiffuseFlows', 'DiffuseFlows']
X = df[features]
y = df['PowerConsumption_Zone1']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestRegressor(n_estimators=100, random_state=42)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

joblib.dump(model, 'model.pkl')
```

app.py

This file is the Streamlit web application. It loads the saved model.pkl, presents five input fields to the user, and displays the predicted Zone 1 power consumption when the Predict Consumption button is clicked.

```
import streamlit as st
import pandas as pd
import joblib

model = joblib.load('model.pkl')
st.title('Electricity Consumption Prediction')

temperature      = st.number_input('Temperature (C)',    value=18.0)
humidity          = st.number_input('Humidity (%)',      value=68.0)
wind_speed        = st.number_input('Wind Speed',        value=2.0)
general_diffuse_flows = st.number_input('General Diffuse Flows', value=183.0)
diffuse_flows      = st.number_input('Diffuse Flows',     value=75.0)

if st.button('Predict Consumption'):
    input_data = pd.DataFrame([[temperature, humidity, wind_speed,
                                general_diffuse_flows, diffuse_flows]],
                               columns=['Temperature', 'Humidity', 'WindSpeed',
                                         'GeneralDiffuseFlows', 'DiffuseFlows'])
    prediction = model.predict(input_data)
    st.success(f'Predicted Zone 1 Power Consumption: {round(prediction[0],2)} watts')
```

3. User Interface / Application

The user interface was built using Streamlit, a Python library that creates simple web applications without requiring HTML, CSS, or JavaScript. The application runs locally in the browser at <http://localhost:8501>.

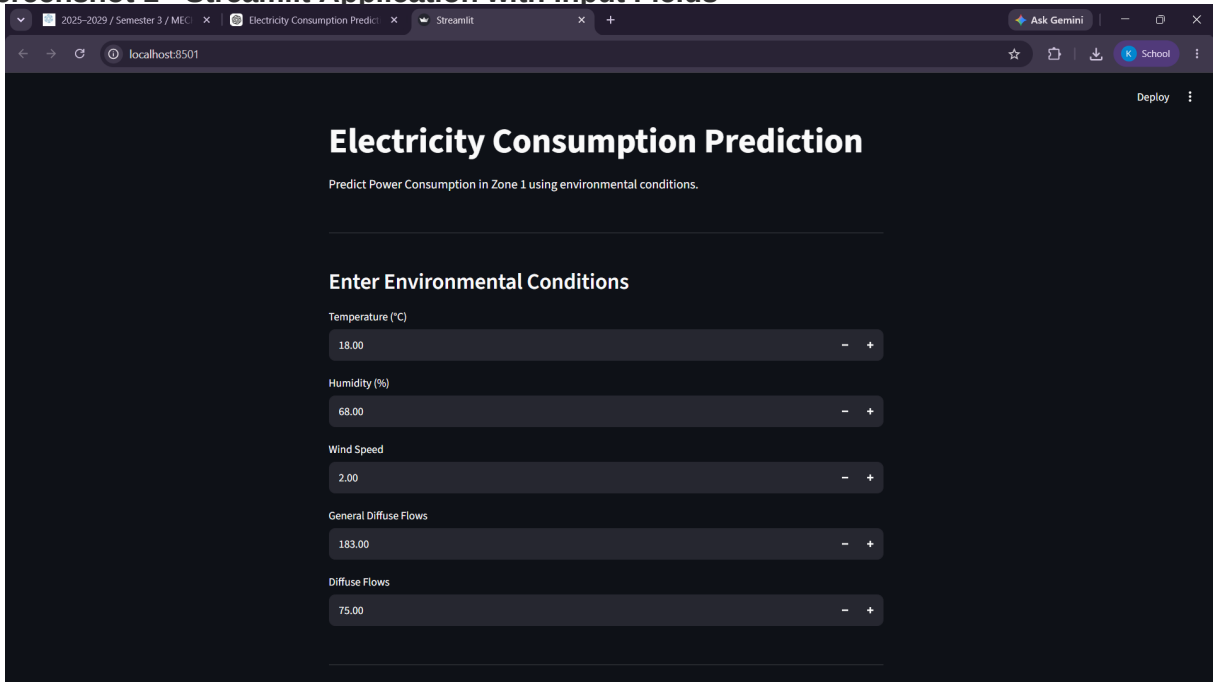
How to Launch

```
Step 1: pip install -r requirements.txt
Step 2: python train_model.py
Step 3: python -m streamlit run app.py
Step 4: Open http://localhost:8501 in your browser
```

The application consists of a single page with a title, a short description, five number input fields, a Predict Consumption button, and a result display area. It runs fully offline and does not require any database or internet connection.

4. Input and Output Screenshots

Screenshot 1 - Streamlit Application with Input Fields

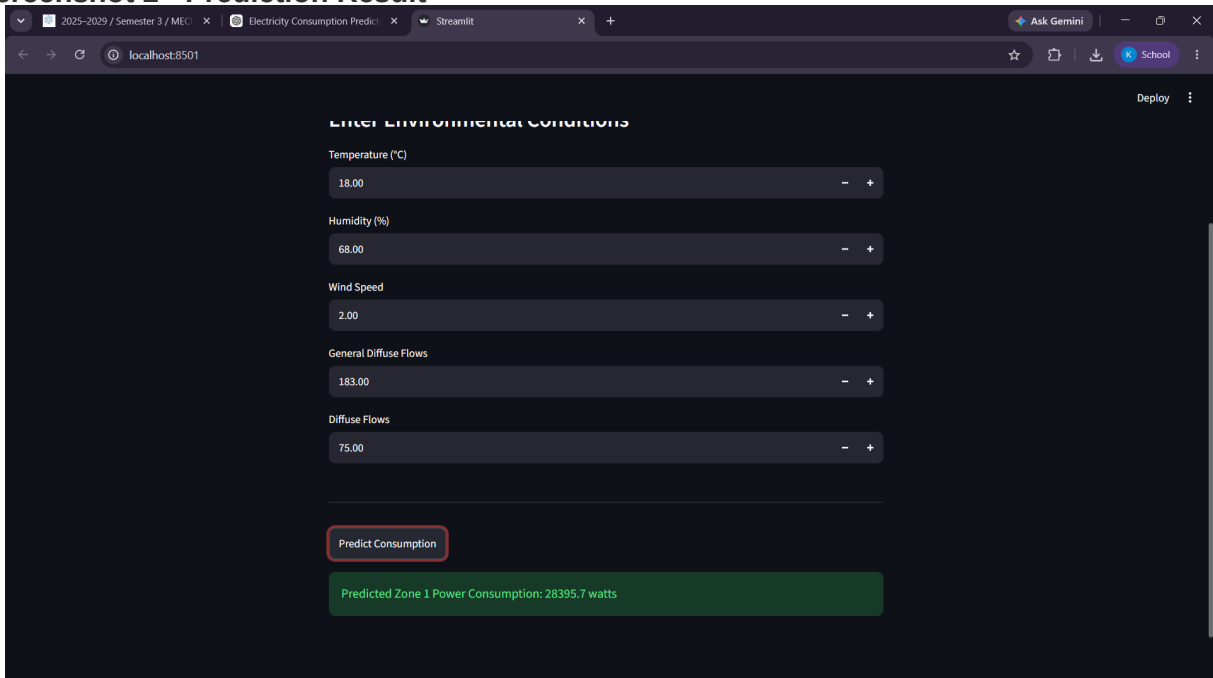


The screenshot shows a web browser window with the URL `localhost:8501`. The page title is "Electricity Consumption Prediction". Below the title, it says "Predict Power Consumption in Zone 1 using environmental conditions." The main section is titled "Enter Environmental Conditions" and contains five input fields, each with a sample value and a range indicator (min and max):

Parameter	Value	Range
Temperature (°C)	18.00	- +
Humidity (%)	68.00	- +
Wind Speed	2.00	- +
General Diffuse Flows	183.00	- +
Diffuse Flows	75.00	- +

Fig 1: Streamlit app showing the five input fields with sample values

Screenshot 2 - Prediction Result



The screenshot shows the same web browser window as Screenshot 1, but with the "Predict Consumption" button highlighted by a red box. Below the button, a green box displays the prediction result: "Predicted Zone 1 Power Consumption: 28395.7 watts".

Fig 2: Predicted Zone 1 Power Consumption displayed after clicking the button

Screenshot 3 - Model Evaluation Results (Terminal)

```
PS D:\E\Project\IBM> cd D:\E\Project\IBM\Electricity_Consumption_Prediction
>>
PS D:\E\Project\IBM\Electricity_Consumption_Prediction> python train_model.py
>>
Dataset loaded successfully!
Total rows: 52416

Training samples: 41932
Testing samples : 10484

Training the Random Forest model... (this may take a moment)
Model training complete!

-----
MODEL EVALUATION RESULTS
-----
MAE   : 3214.31
MSE   : 21522596.95
RMSE  : 4639.25
R2    : 0.5738
-----

Model saved as model.pkl
Done! You can now run: streamlit run app.py
PS D:\E\Project\IBM\Electricity_Consumption_Prediction>
PS D:\E\Project\IBM\Electricity_Consumption_Prediction>
PS D:\E\Project\IBM\Electricity_Consumption_Prediction> █
```

Fig 3: Terminal output showing MAE, MSE, RMSE and R2 Score

Screenshot 4 - Actual vs Predicted Graph

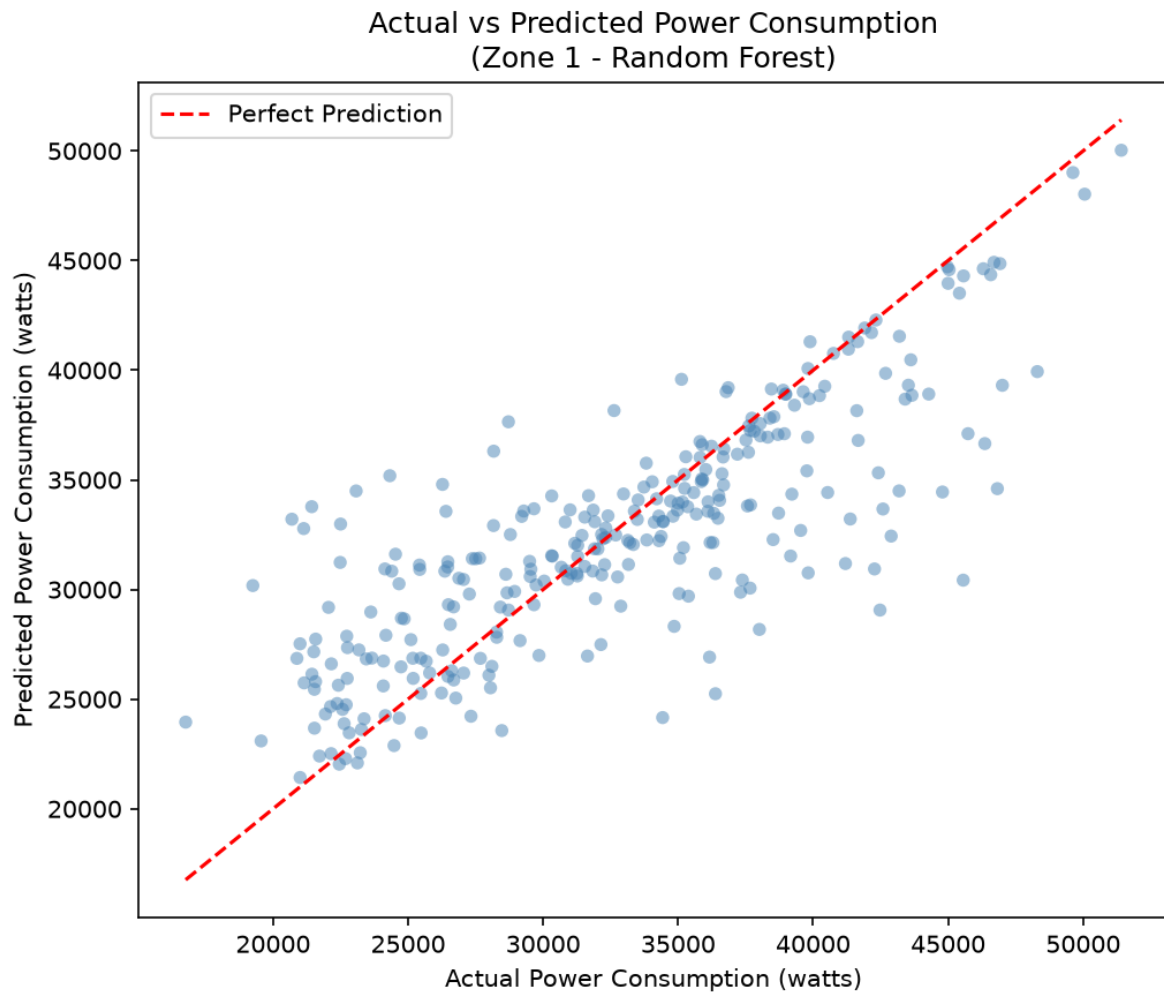


Fig 4: Actual vs Predicted scatter plot

5. Final Results

Phase 4 - Final Model Evaluation

Metric	Value
Mean Absolute Error (MAE)	3214.31 watts
Mean Squared Error (MSE)	21,522,596.95
Root Mean Squared Error (RMSE)	4639.25 watts
R2 Score	0.5738

Comparison with Phase 3 Models

Model	MAE	RMSE	R2 Score
Linear Regression	5185.74	6312.08	0.21
Gradient Boosting	4512.90	5747.29	0.35
Random Forest (Phase 3)	3426.65	4795.16	0.54
Random Forest (Phase 4-Final)	3214.31	4639.25	0.5738

Model Performance Comparison Graph

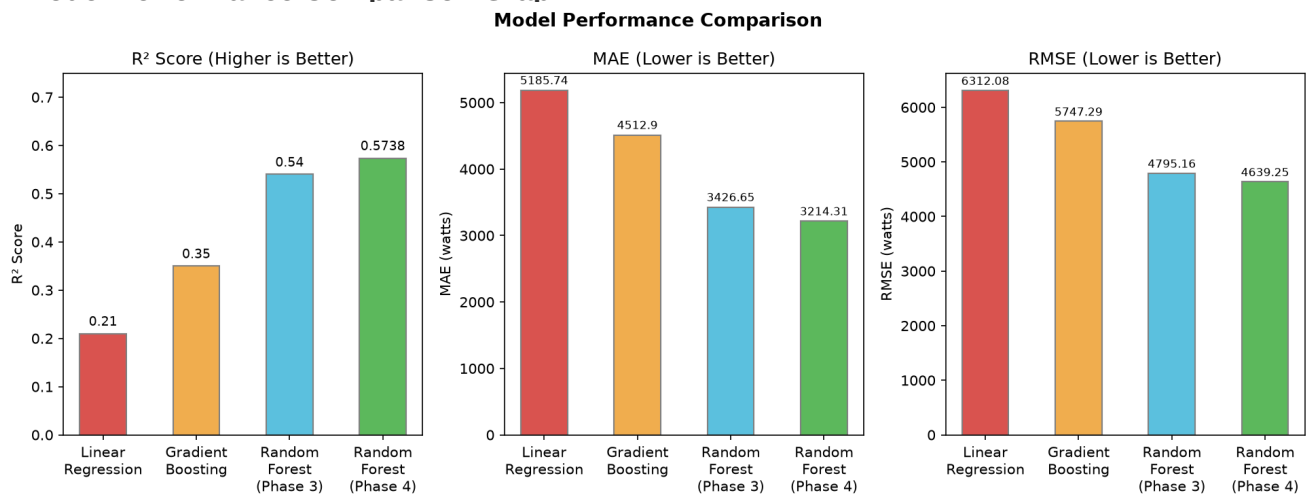


Fig 5: Bar chart comparing all models across MAE, RMSE and R2 Score

Actual vs Predicted Graph

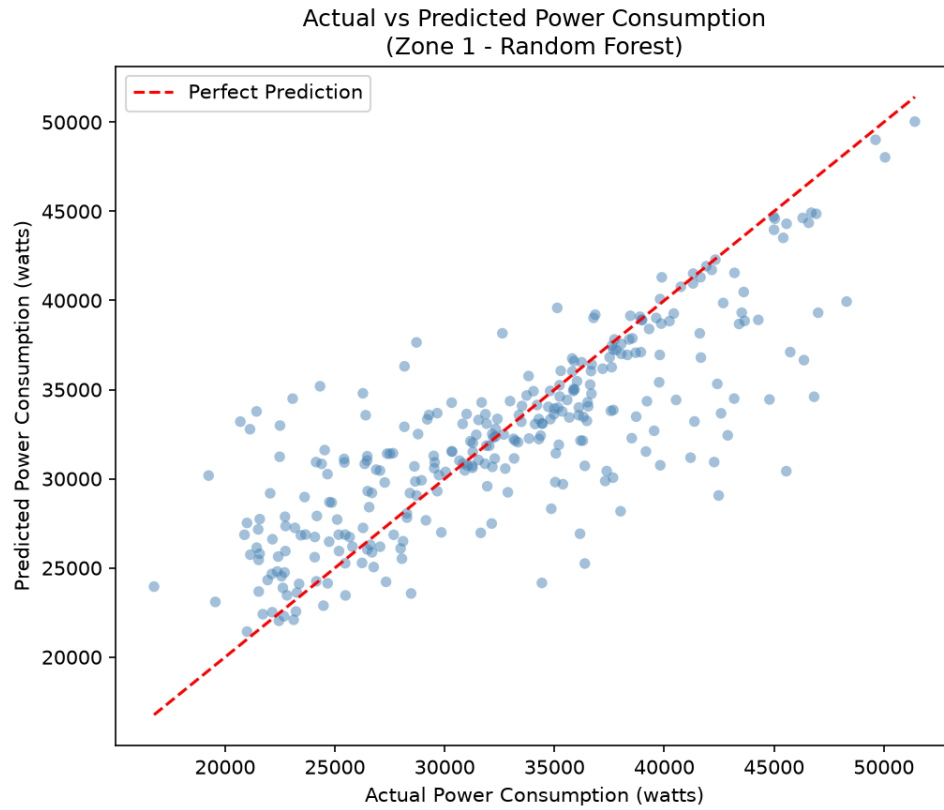


Fig 6: Scatter plot of Actual vs Predicted Zone 1 power consumption

Feature Importance Graph

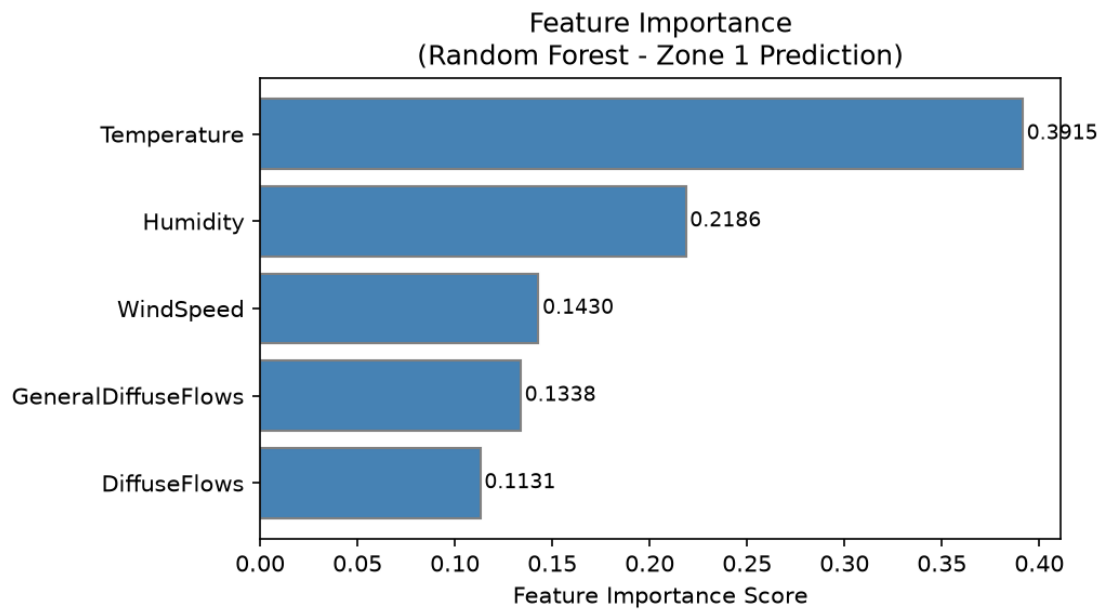


Fig 7: Feature importance scores from the Random Forest model

Feature Importance Summary

Rank	Feature	Importance
1	Temperature	Highest
2	Humidity	Second

3	GeneralDiffuseFlows	Moderate
4	DiffuseFlows	Moderate
5	WindSpeed	Lowest

6. Conclusion

In this project, a machine learning model was developed to predict electricity consumption in Zone 1 of a residential area using five environmental features: Temperature, Humidity, WindSpeed, GeneralDiffuseFlows, and DiffuseFlows.

The problem was treated as a regression task because the target variable, PowerConsumption_Zone1, is a continuous numerical value. Three regression algorithms were compared in Phase 3. Random Forest Regression performed best across all evaluation metrics and was selected as the final model.

The final Random Forest model was trained on 80% of the dataset (41,932 rows) and evaluated on the remaining 20% (10,484 rows). It achieved a MAE of 3214.31, RMSE of 4639.25, and an R2 Score of 0.5738, meaning the model explains approximately 57.4% of the variation in Zone 1 electricity consumption.

A simple web application was developed using Streamlit. The application allows a user to enter five environmental values and click a button to receive the predicted electricity consumption for Zone 1 instantly. Overall, this project demonstrates how machine learning can be applied to a real-world electricity dataset to build a functional prediction system.

7. Project Limitations

- Moderate accuracy: The model achieved an R2 Score of 0.5738, meaning it explains about 57% of the variation. The remaining 43% is influenced by other factors not present in the dataset.
- Only Zone 1 is predicted: The model covers only Zone 1. Zones 2 and 3 are not included.
- Environmental features only: The model uses only weather-related features. Time of day, day of week, holidays, or number of occupants are not included.
- Dataset is from one region and year: The dataset contains readings from one location for 2017. It may not perform equally on data from other regions or periods.
- No real-time data: The application does not fetch live weather data. The user must manually enter values each time.

8. Future Enhancements

- Add time-based features: Including hour of day, day of week, or month could improve predictions since electricity usage follows daily and seasonal patterns.
- Predict all three zones: The model could be extended to predict PowerConsumption_Zone2 and PowerConsumption_Zone3 as well.
- Improve model accuracy: Hyperparameter tuning (adjusting number of trees, depth, etc.) could be applied to improve the R2 Score.
- Use a larger and more recent dataset: Training on data from different years and regions would make the model more general and reliable.
- Display graphs inside the Streamlit app: The Actual vs Predicted plot and Feature Importance chart could be embedded directly inside the application.

9. GitHub Repository

The complete project code is available on GitHub. The repository contains all source files required to reproduce the results.

Repository Contents

File	Description
powerconsumption.csv	Dataset (52,416 rows)
train_model.py	Model training and saving script
app.py	Streamlit web application
generate_graphs.py	Result graph generation script
requirements.txt	Python dependencies
README.md	Project documentation and run instructions

How to Run from GitHub

```
git clone https://github.com/<your-username>/Electricity-Consumption-Prediction.git
cd Electricity-Consumption-Prediction
pip install -r requirements.txt
python train_model.py
python -m streamlit run app.py
```

Note: model.pkl is not included in the repository (363 MB). Run train_model.py to generate it locally after cloning.